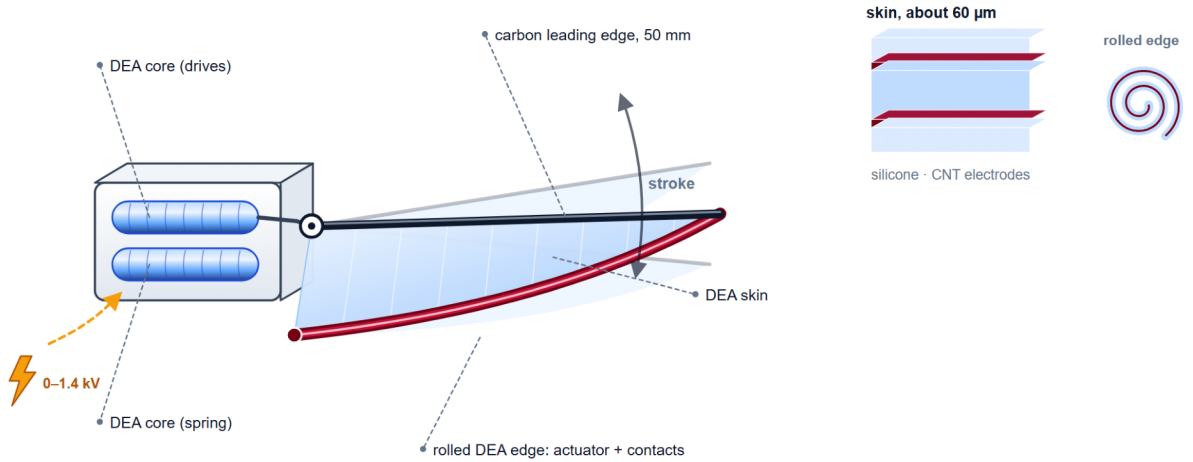
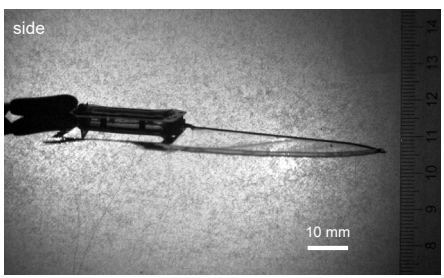
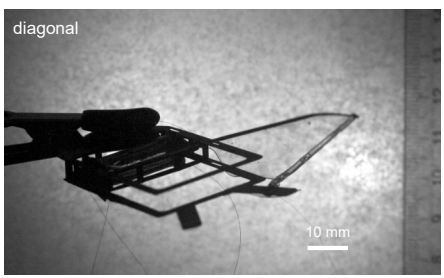
**a**


DEA core (drives)
carbon leading edge, 50 mm
stroke
DEA skin
0–1.4 kV
DEA core (spring)
rolled DEA edge: actuator + contacts
skin, about 60 µm
rolled edge
silicone · CNT electrodes

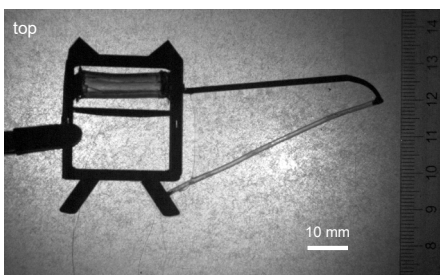
**b**


side
10 mm

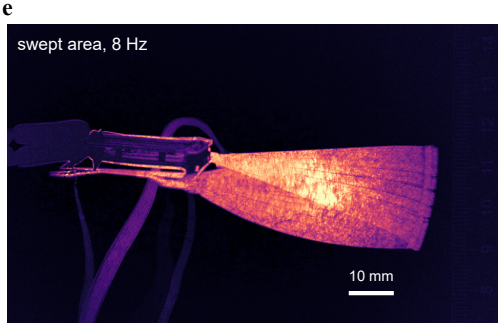
**c**


diagonal
10 mm

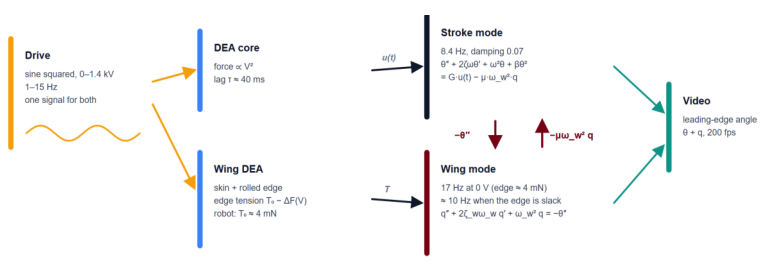
**d**


top
10 mm

**e**


swept area, 8 Hz
10 mm

**f**


Drive
sine squared, 0–1.4 kV
1–15 Hz
one signal for both
DEA core
force ∝ V²
lag τ ≈ 40 ms
Wing DEA
skin + rolled edge
edge tension T₀ − ΔF(V)
robot: T₀ ≈ 4 mN
u(t)
Stroke mode
8.4 Hz, damping 0.07
θ'' + 2ζω₀θ' + ω₀²θ + βθ³
= G·u(t) − µ·ω_w²·q
−θ''
−µω_w² q
τ
Wing mode
17 Hz at 0 V (edge ≈ 4 mN)
≈ 10 Hz when the edge is slack
q'' + 2ζ_w ω_w q' + ω_w² q = −θ''
Video
leading-edge angle
θ + q, 200 fps
voltage slackens the edge → the wing mode leaves the band → the stroke is freed